

# Control Structures (If-Else, Loops, Switch)

## 1. Decision Making (If-Else)

Used to make decisions in a program based on conditions.

### Simple If

Syntax:

```
if(condition) { // code to execute if true }
```

If there is single statement in if block no need for curly brackets :

```
If(a>0)
```

```
    Printf("%d is positive",a);
```

### If-Else

Syntax:

```
if(condition) { // if true } else { // if false }
```

### Nested If-Else

Syntax:

```
if(condition1) { ... } else if(condition2) { ... } else { ... }
```

## 2. Conditional Operator (Ternary)

Shorthand for if-else. Syntax: condition ? expression1 : expression2;

Example: `int max = (a > b) ? a : b;`

## 3. Switch Case

Used to select one of many code blocks to be executed.

Syntax:

```
switch(expression) {  
    case constant1: // code; break;  
    case constant2: // code; break;
```

```
default: // code;  
}
```

## 4. Loops

Loops are used to repeat a block of code.

### Types of Loops

**1. While Loop (Entry Controlled):** Checks condition before execution.

Syntax: `while(condition) { // code; increment; }`

**2. For Loop (Entry Controlled):** Preferred when number of iterations is known.

Syntax: `for(initialization; condition; increment) { // code; }`

**3. Do-While Loop (Exit Controlled):** Executes at least once.

Syntax: `do { // code; } while(condition);`

### When to use which loop?

- **For Loop:** Use when you know exactly how many times the loop should run (e.g., iterating through an array).
- **While Loop:** Use when the number of iterations depends on a condition that might change (e.g., reading until a user types "exit").
- **Do-While Loop:** Use when the block of code must run at least once, regardless of the condition (e.g., displaying a menu).